

Syif M. Bhuiyan

Dhaka, Bangladesh | +880 183 191 3487 | syifnibir@gmail.com

Portfolio: syifbhuiyan.github.io | LinkedIn: linkedin.com/in/syif-m-bhuiyan-a7250a119

RESEARCH INTERESTS

Primary: Agricultural & Resource Economics, Agricultural Finance & Credit Risk, Agri-Fintech.

Secondary: Applied Econometrics, Land Use & Spatial Economics, Climate Variances.

EDUCATION

Bachelor of Science (B.Sc.) in Economics & Finance | Southeast Missouri State University, Cape Girardeau, MO

Economics Major GPA: 3.27 / 4.00

Quantitative Coursework: Macroeconomic Theory (A), Business Statistics I (A), Healthcare Economics (A), Applied Calculus (B), Microeconomic Theory (B), Managerial Economics (B).

Bachelor of Business Administration (BBA) Coursework | Institute of Business Administration (IBA), University of Dhaka, Bangladesh

Completed 30 credit hours of core business, accounting, and quantitative coursework at the nation's premier business institution before transferring to the US.

CONFERENCE PRESENTATIONS

Missouri Valley Economic Association (MVEA) Annual Conference

- Selected by faculty to present empirical research findings to regional economists and academic peers in St. Louis, MO.

EMPIRICAL RESEARCH & DATA PROJECTS

Panel Data Analysis of Climate Variances on Regional Crop Yields | Independent Researcher

- Constructed a data pipeline using Python (Pandas) to merge county-level crop yield data (USDA NASS) with weather metrics (NOAA), utilizing the FCC API for reverse-geocoding misaligned spatial coordinates.
- Estimated a two-way fixed-effects panel regression (linearmodels.PanelOLS) to isolate the causal impact of temperature and precipitation variances on agricultural outputs, controlling for unobserved local traits and macroeconomic shocks.
- Revealed severe omitted variable bias (Simpson's Paradox) in the raw data; after controlling for spatial and temporal fixed effects (F-test $P=0.0014$), the model isolated a positive temperature coefficient (2.3979).

The Airbnb Effect: Impact on NYC Housing Markets | Independent Researcher

- Merged short-term rental listings (Inside Airbnb) with neighborhood housing indices (Zillow ZHVI) across 164 NYC ZIP codes using Python to conduct spatial economic analysis.
- Executed an Ordinary Least Squares (OLS) regression (statsmodels) to evaluate the impact of rental density on localized housing prices.

- Identified a statistically significant relationship ($P < 0.001$) where each additional listing correlated with a \$585 increase in annual home values ($R\text{-squared} = 0.097$).

Algorithmic Fairness in Digital Credit Risk | Independent Researcher

- Audited fintech credit scoring models for demographic parity using the Home Credit Default Risk dataset, focusing on equitable financial inclusion.
- Recalibrated a baseline Logistic Regression model using Microsoft's Fairlearn toolkit and scikit-learn.
- Equalized the "High Risk" selection rate to 41.6% across demographic groups via threshold optimization, mitigating a baseline disparity of 90.5% vs. 14.0%.

PROFESSIONAL EXPERIENCE

Project Manager (Agri-Fintech) | WeGro Technologies Ltd. *17 months*

- Authored and secured grant proposals to fund data-driven agricultural financing and farmer credit initiatives.
- Integrated disparate agricultural production data with financial metrics using SQL and Python to refine farmer credit-scoring parameters.
- Built interactive Power BI dashboards to track project KPIs and financial disbursement metrics.

Remote AI Data Analyst (Contract) | Telus International AI & Toloka *10 months | Ongoing*

- Evaluated generative AI outputs using Reinforcement Learning from Human Feedback (RLHF) pipelines.
- Audited edge-case model responses for logical coherence and algorithmic bias.

Marketing & Sales Analytics Internships | MetLife & British American Tobacco *6 months combined*

- Automated weekly sales reporting workflows by combining Python scripts, SQL queries, and Excel VBA.

TECHNICAL SKILLS

Languages & APIs: Python (pandas, numpy, linearmodels, statsmodels, scikit-learn), R, SQL, REST APIs (FCC).

Econometrics: OLS Regression, Two-Way Fixed-Effects Panel Data, Logistic Regression, Causal Inference, Hypothesis Testing.

Tools & BI: Tableau, Power BI, Advanced Excel (VBA), Git/GitHub.

Certifications: Google Advanced Data Analytics Professional, DataCamp Data Analyst Associate.

UNIVERSITY LEADERSHIP & SERVICE

President, Bangladeshi Students' Association (BSA) | Southeast Missouri State University

- Directed organizational budgeting, event planning, and cultural outreach for the university's international student body.

International Student Ambassador

- Appointed by university administration to mentor incoming international cohorts and represent the university at official campus events.